

1 Introduction and Dataset Overview

The goal of this project is to forecast 600 different time series, each with its own forecast horizon h . The series vary in length and frequency, so we do not expect a single model to perform best everywhere. Forecast accuracy is measured using the Mean Absolute Error (MAE) in two settings: cross-validation, where we predict the last h points of each series, and the Kaggle leaderboard, which evaluates our forecasts on a hidden test set.

We implemented three main model families: ARIMA, TBATS, and TSLM with time as a covariate. For each one, we first ran a default baseline, then applied the same cross-validation scheme across all 600 series and compared the average MAE. We then tuned key hyperparameters and examined how they affected the MAE, using residual diagnostics on selected “best” and “worst” series to understand when each model succeeds or fails. Finally, motivated by the heterogeneity of the data, we built a hybrid forecasting strategy that chooses a model per series based on its length, frequency, and apparent complexity, aiming to improve Kaggle performance while retaining strong cross-validated accuracy.

2 Forecasting Methods

2.1 ARIMA

The first forecasting method we used was ARIMA. According to the results obtained on our time series, it performs best on time series that show consistent and repeated patterns: long series with stable seasonality but also a clear and steady evolution over time.

In these situations, the structure is strong enough for ARIMA to learn how past values relate to future ones and to effectively capture the seasonal parts of the time series. In practice, ARIMA works well when the behaviour of the series changes smoothly, without sudden jumps or irregular fluctuations. However, it struggles on very short time series, as well as on series with more complex seasonality or other kinds of irregular patterns. Using cross-validation on the 600 time series, the best ARIMA configuration (stepwise = FALSE, approximation = FALSE) achieved an internal average MAE of 554.0935, and a score of 565.6687 when submitted to Kaggle.

Overall, ARIMA is a solid and reliable model for well-structured series, but its performance drops significantly on short or highly irregular data.

2.2 Linear Regression (TSLM)

The second forecasting method we used was time-series linear regression (TSLM). We included it as a simple, interpretable model that should work best when a series can be well described by a straight-line trend and stable, repeating seasonality. In these cases, trend and seasonal dummy terms can capture most of the structure with very few parameters. However, TSLM cannot adapt to non-linear growth or changing patterns, so the residuals often remain autocorrelated and the forecasts miss turning points. Using cross-validation on the 600 series, the trend-only model reached an average MAE of 1003.5006, the trend + season model improved this slightly to 995.1405, and the Box-Cox variant performed worse with 1286.8247. The Kaggle MAE score was 970.44.

Overall, TSLM is a useful and fast baseline, but not suitable as a final forecasting method for this dataset.

2.3 TBATS

The third forecasting method applied in this project was TBATS, a model specifically designed to accommodate complex, multiple, and non-integer seasonal patterns. Given the heterogeneous nature of the dataset—comprising hourly, daily, weekly, monthly, and quarterly time

series with varying lengths—TBATS is particularly appropriate, as it can flexibly model seasonal structures that traditional approaches may fail to capture.

In our cross-validation exercise, the default TBATS configuration produced the strongest overall performance, achieving a mean MAE of 477.5424, considerably outperforming both ARIMA and TSLM. To investigate whether accuracy could be improved further, we tested several hyperparameter adjustments, including enforcing a Box–Cox transformation, removing the trend component, and introducing a damped trend. Contrary to our expectations, these modifications did not enhance forecast accuracy, and all performed slightly worse than the default model, indicating that TBATS’ automatic model selection was already near-optimal for this dataset. When trained on the full series and evaluated on the Kaggle leaderboard, TBATS achieved a competitive score of 558.40071, confirming its suitability for longer series with pronounced or complex seasonal behaviour, though it remains less reliable for very short or highly noisy time series.

	ARIMA	TBATS	Linear Regression
Strengths	Flexible, adapts to different models with different parameters	Can handle complex and multiple seasonalities in the data	Works well when trend/seasonality are stable and linear
Weaknesses	Fails with multiple seasonal patterns	Can overfit seasonal components, making the model sensitive to noise	Too rigid for non-linear patterns due to linear relationship assumptions
Kaggle score	561.686	558.40071	970.43957
CV MAE	557.2463	477.5424	1003.501

Table 1: Summary of strengths, weaknesses, and model performance.

3 Final Solution

Our final model used a decision tree to select forecasting methods based on time series length, seasonality complexity, and frequency. For short, simple series (length < 3), the naive method was applied due to insufficient data for complex models. Long, complex series with multiple seasonal patterns used TBATS, as it performs best in such cases, but may fall back to seasonal naive to avoid overfitting or convergence issues with noisy data. Simpler series were modeled using Auto ARIMA, which adapts well to non-multiple seasonality, or ETS, chosen via the lowest AICc, with naive and seasonal naive as fallbacks. Linear regression was excluded due to lower cross-validation and Kaggle performance.

This approach achieved a Kaggle score of 535.18389, indicating room for improvement. A brute-force method applying all models to each series could potentially improve results but at the cost of runtime and storage.

4 Conclusion

Overall, TBATS was the strongest single method: it achieved the lowest cross-validation MAE and a competitive Kaggle score, especially on long, seasonal series. ARIMA performed reliably on well-structured series but struggled with short or highly irregular ones, while TSLM remained clearly less accurate than both. Our final hybrid approach, which chooses between TBATS, ARIMA/ETS, and naive variants based on each series’ length and seasonality, delivered our best Kaggle performance but showed some loss of generalisation on the private leaderboard. A more exhaustive per-series model comparison (fitting several candidate models and selecting the one with the lowest validation MAE) is a natural next step to make the forecasts more robust.